

Project Report

Student Management System (Java Swing Desktop Application)

Applicant Name:	Jay Kumar
Application ID:	KTS020260715548
Internship Domain:	Java Development Internship
Organization:	Kinetrexa Software Private Limited
Task:	Task 1 - Student Management System
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1. Objective

The objective of this project is to develop a Java-based desktop application that allows efficient management of student records. The system applies core Object-Oriented Programming principles and supports adding, updating, deleting, searching, and displaying student information through a graphical user interface.

2. Technology Stack

- Language: Java (JDK 8+)
- GUI Framework: Java Swing (JFrame, JTable, GridBagLayout)
- Data Storage: CSV text file (data/students.csv)
- Build/Run: javac / java (no external dependencies)

3. System Architecture

The application follows a modular, layered architecture that separates data, logic, and presentation concerns:

- **model** package - the Student class, representing a single student record.
- **dao** package - StudentDAO, which handles all CSV file reading/writing and CRUD logic.
- **exception** package - custom checked exceptions (InvalidStudentDataException, StudentNotFoundException) for input validation and error handling.
- **gui** package - StudentManagementGUI, the Swing interface that wires user actions to the DAO layer.
- **Main** - application entry point that launches the GUI on the Swing event thread.

4. Key Features Implemented

- Student CRUD Operations - add, update, and delete student records via the GUI.
- Object-Oriented Programming - encapsulated Student model with clear class responsibilities.
- File Handling - persistent storage using a CSV file, loaded and saved through StudentDAO.

- Exception Handling - custom exceptions for invalid input and missing records, plus try-catch blocks around all file operations.
- Search & Filter - keyword search across ID, name, and course fields.
- Modular Code Structure - code organized into model, dao, exception, and gui packages.

5. Validation Rules

- Name and Course are required fields.
- Age must be a number between 1 and 120.
- Email, if provided, must match a standard email format.
- Phone, if provided, must contain 7 to 15 digits.
- Fields cannot contain commas, since the CSV format uses commas as the field delimiter.

6. How to Run

1. Compile: `javac -d bin $(find src -name "*.java")`

2. Run: `java -cp bin Main`

The application window opens and data.csv is created automatically on first run.

7. Conclusion

This project demonstrates the practical application of Java fundamentals, Object-Oriented Programming, file handling, exception handling, and GUI development using Swing. The modular design makes the codebase easy to extend, for example by replacing CSV storage with a database in a future iteration.